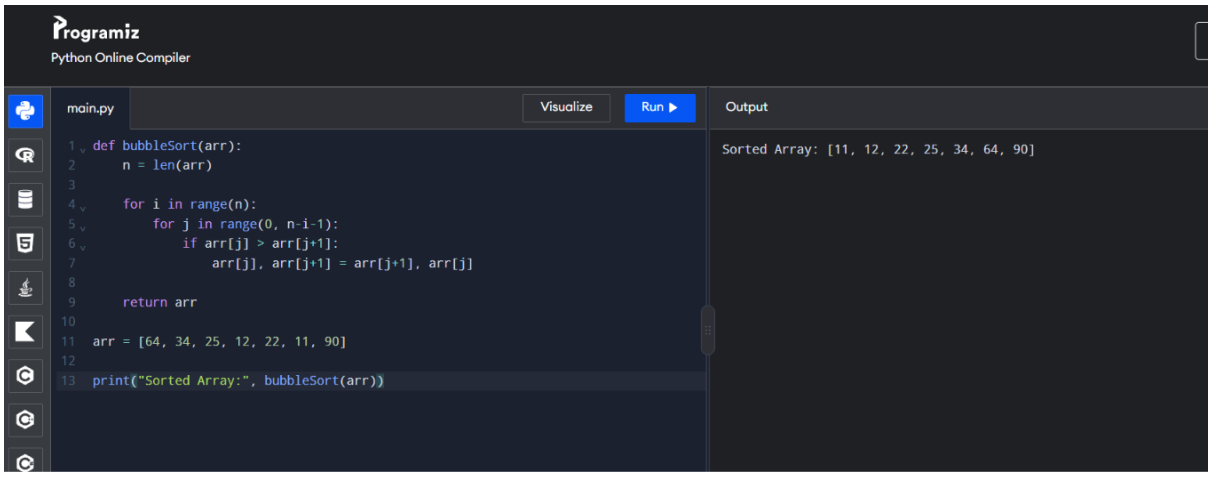
3. Bubble Sort with Big-O Analysis

Aim

To sort the given array in ascending order using the Bubble Sort algorithm and analyze its time complexity.

Procedure

1. Read the array elements.
2. Compare adjacent elements in the array.
3. Swap the elements if they are in the wrong order.
4. Repeat the process for all elements until the array is sorted.
5. Display the sorted array.
6. Analyze the time complexity of the algorithm.



Result

Thus, the given array was sorted successfully using the Bubble Sort algorithm.

Time Complexity:

- Best Case: O(n)
- Average Case: $O(n^2)$
- Worst Case: $O(n^2)$